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NAAC
GRADE A+
ACCREDITED UNIVERSITY

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POWER BI RETAIL SALES PERFORMANCE REPORT

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Introduction

Retail organisations generate large volumes of transaction data every day. However, raw data stored across different tables does not immediately provide useful business information. Business intelligence tools such as Power BI help convert this data into dashboards that can be understood quickly by managers and analysts. In this project, a retail dataset was organised into related tables for sales, customers, products, stores and calendar details. These tables were used to build a three-page report with interactive charts and filters.

Objectives of the Report

- To create a centralized retail business dashboard using Microsoft Power BI.
- To analyse overall sales, quantity, customer count and order count using KPI cards.
- To compare sales performance across products, categories and cities.
- To understand customer behaviour using gender, age group and customer segment analysis.
- To identify preferred payment methods and compare store types.
- To make the dashboard interactive using slicers and filters.
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Data Model and Dashboard Design

The report uses multiple related tables visible in the Power BI Data pane: Calendar, CustomerDetails, retailb products, retailb sales and retailb stores. The sales table serves as the main transaction source, while the remaining tables provide descriptive information required for analysis. This structure supports the creation of visuals across different business dimensions. The

dashboard was divided into three pages so that each page has a clear analytical purpose and does not become overcrowded.

Key Measures and Analysis Areas

Measure / Area	Purpose
Total Sales	Shows the overall revenue generated from retail transactions.
Total Quantity	Summarises the quantity of products included in the analysis.
Total Customers	Shows the number of customers represented in the dashboard.
Total Orders	Shows the total order activity available in the report.
Product and Category Analysis	Identifies high-performing products and product groups.
Customer Analysis	Examines gender, age group and customer segment behaviour.
Store Analysis	Compares orders and performance by store type.

Dashboard Page 1 – Overall Sales and Product Performance

The first dashboard page presents the overall performance of the retail business. Four KPI cards provide a quick summary of the most important measures: Total Sales of 460K, Total Quantity of 97, Total Customers of 16 and Total Orders of 16. This top-level summary enables users to understand the current scale of business activity before moving to detailed charts.

The page also contains a category slicer and a Month-Year slicer. These controls allow users to focus on selected product categories or time periods. The main visuals analyse Total Sales by Product Name, Total Sales by City and Total Sales by Category. Together, these charts help identify the strongest products, compare geographical sales performance and understand category-level trends.

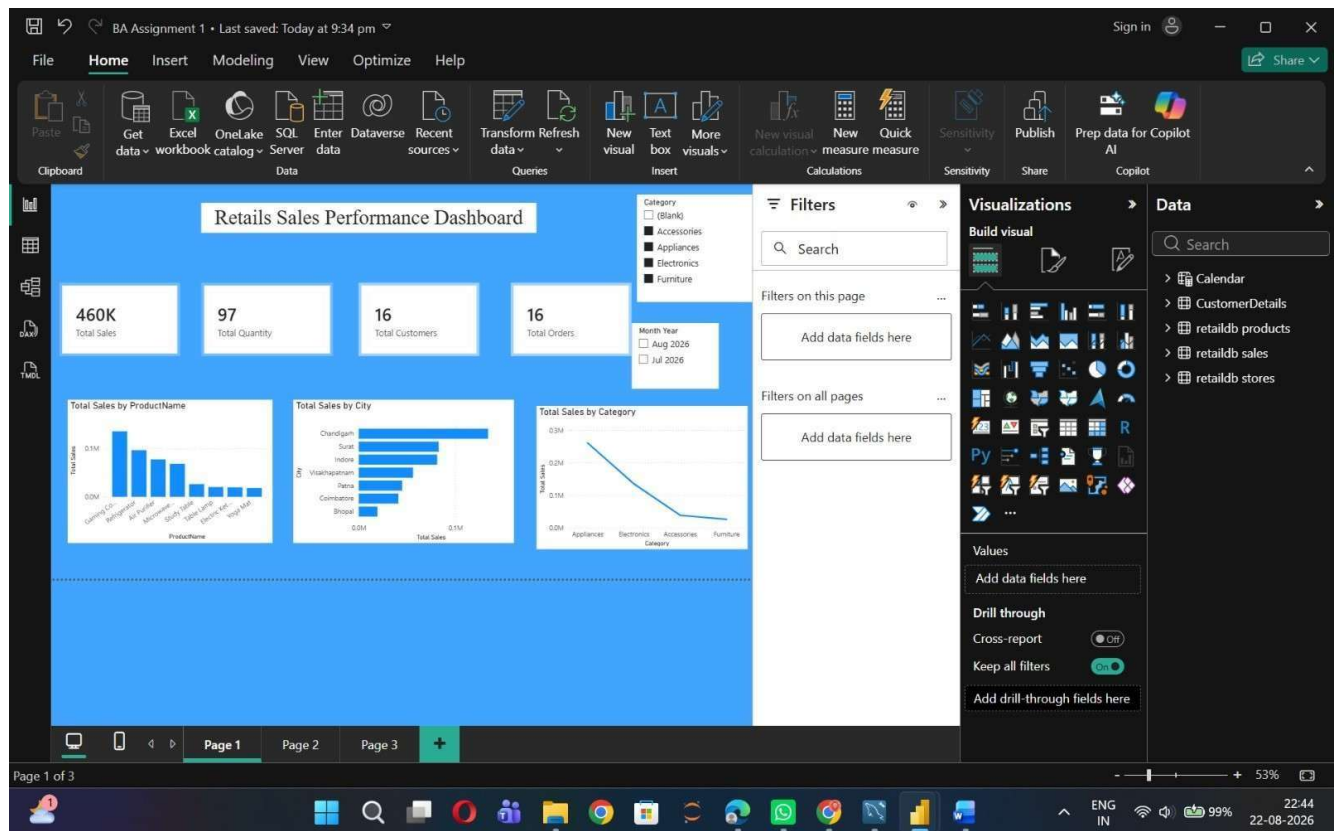


Figure 1: Power BI Dashboard Page 1 – Overall Sales, Product, City and Category Analysis

Key Interpretation

The first page is intended for management-level monitoring. KPI cards provide immediate visibility of business totals, while the supporting charts explain where those sales are coming from. Product and city comparisons can assist in inventory planning, regional sales evaluation and the identification of product categories that require additional marketing attention.

Dashboard Page 2 – Customer and Payment Behaviour

The second dashboard page focuses on customer characteristics and purchasing preferences. A donut chart compares Total Sales by Gender, allowing the analyst to examine how sales are distributed between male and female customers. A horizontal bar chart compares Total Sales by Payment Method, including Credit Card, Cash, UPI, Net Banking and Debit Card.

The page also analyses sales across age groups and customer segments. The age-group chart helps identify the customer population that contributes most strongly to revenue, while the customer-segment visual distinguishes Regular, Premium and New customers. Additional slicers for Gender, Customer Segment and Payment Method make the page interactive and support more focused analysis.

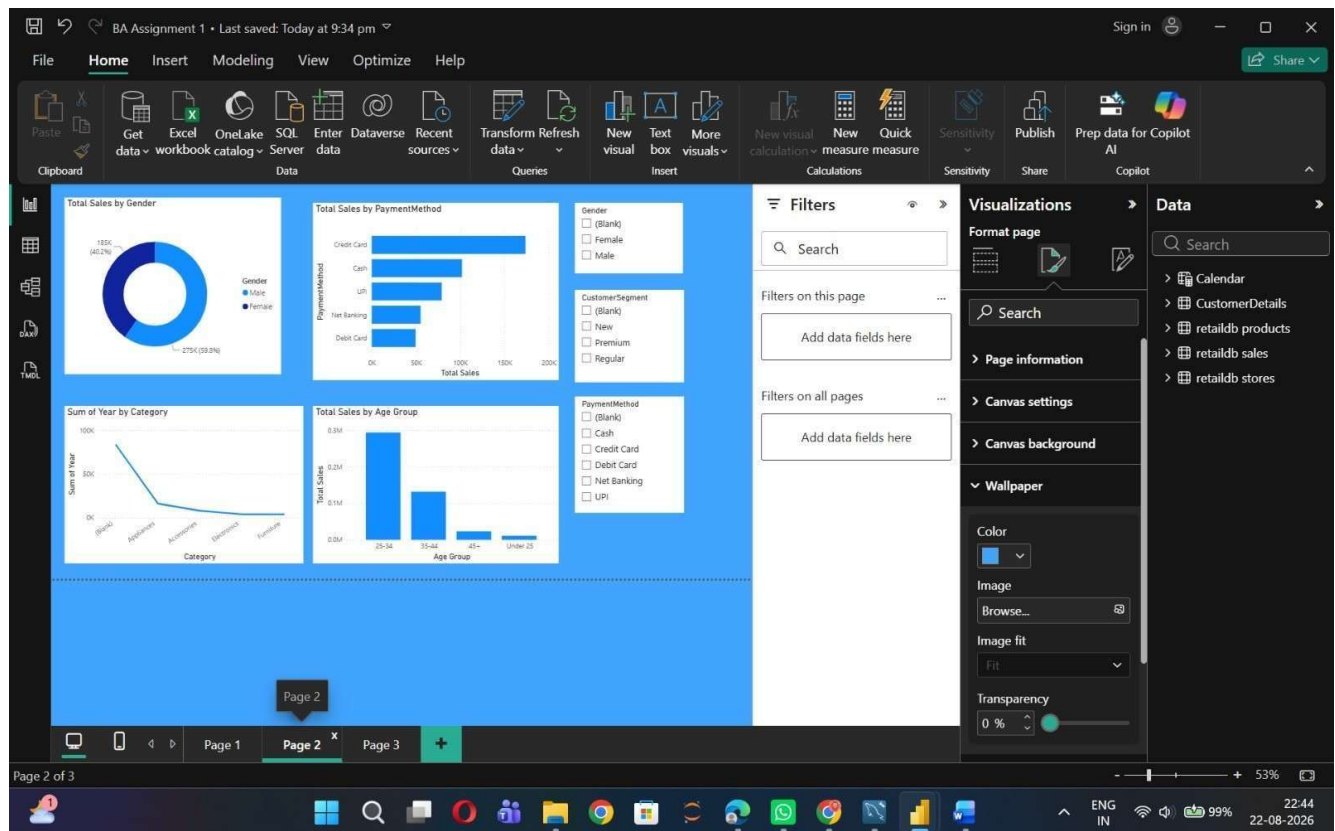


Figure 2: Power BI Dashboard Page 2 – Gender, Payment Method, Age Group and Customer Segment Analysis

Key Interpretation

This page supports customer-oriented decision-making. Understanding the dominant payment methods can help the company improve the checkout experience, while age and segment analysis can support targeted promotions. The interactive slicers make it possible to explore combinations of customer characteristics without creating separate reports for every customer group.

Dashboard Page 3 – Store Type and Customer Segment Performance

The third dashboard page provides a concise comparison of store formats and customer segments. The pie chart displays Total Orders by Store Type, comparing Standalone and Mall stores. This allows the business to understand how order activity is distributed across different retail locations.

A column chart displays Total Sales by Customer Segment. Regular, Premium and New customers are compared to determine which segment contributes most to revenue. This

information can support customer retention programmes, loyalty initiatives and strategies for converting new customers into long-term buyers.

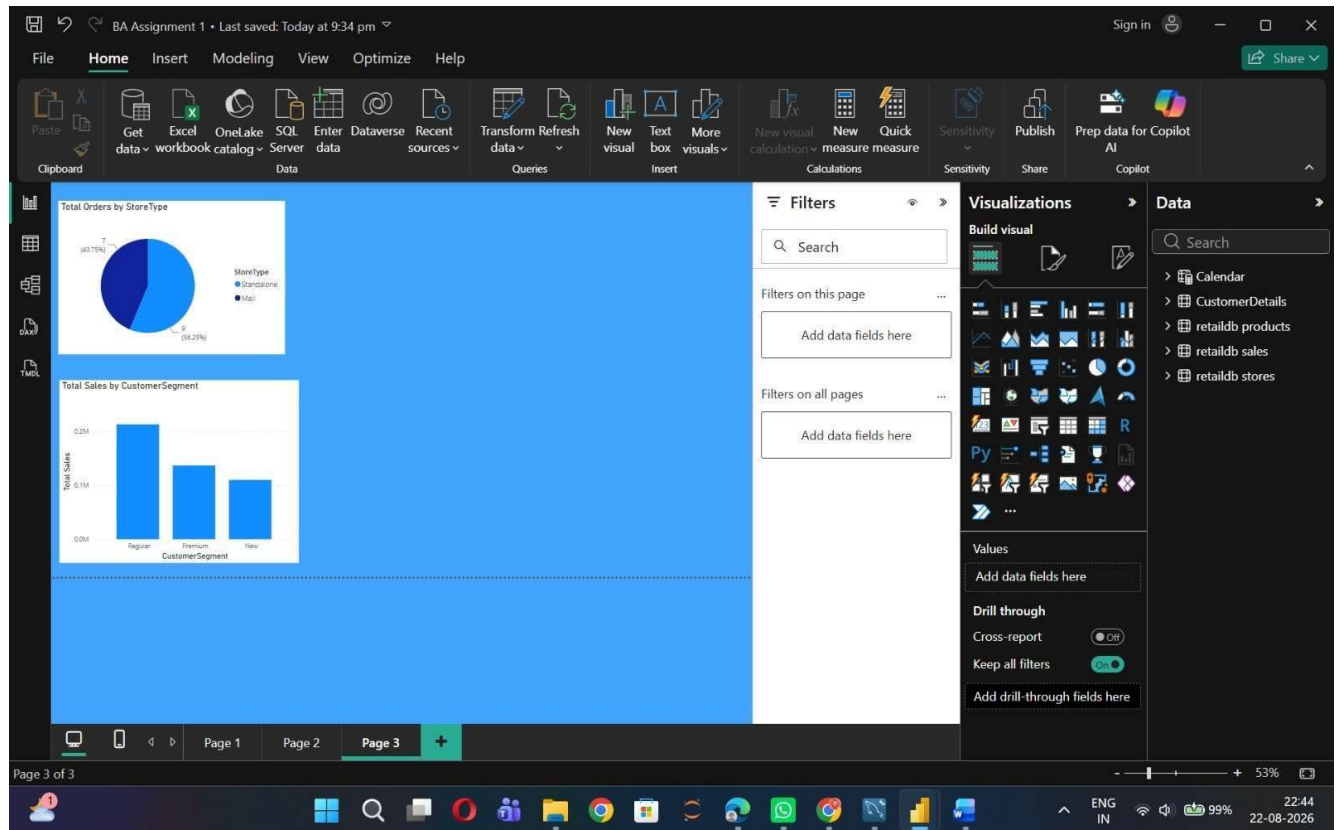


Figure 3: Power BI Dashboard Page 3 – Store Type and Customer Segment Analysis

Key Interpretation

The store-type analysis can help management evaluate the contribution of different retail formats. Customer-segment analysis shows the importance of maintaining relationships with valuable customer groups while also developing strategies to increase the contribution of newer customers.

Overall Findings

- The dashboard provides a single reporting environment for analysing retail business performance.
- KPI cards give a fast summary of sales, quantity, customers and orders.
- Product, category and city visuals provide detailed sales comparisons.
- Customer analysis identifies differences in gender, age group and customer segment behaviour.
- Payment method analysis highlights customer transaction preferences.

- Store-type analysis supports comparison between different retail formats.
- Slicers and filters improve usability by allowing users to perform self-service analysis.

Conclusion

The Power BI Retail Sales Performance Dashboard successfully converts related retail datasets into a structured and interactive business report. By separating the analysis into three pages, the dashboard presents overall performance, customer behaviour and store/customer-segment performance in a clear manner. The combination of KPI cards, charts and slicers enables users to move from high-level business summaries to more detailed analysis. This project demonstrates practical skills in data modelling, dashboard design, visual analytics and business insight generation using Microsoft Power BI.